Practical 4: Gradient Descent to find the local minimum of $y = (x + 3)^2$

In [1]:
```
import numpy as np
import matplotlib.pyplot as plt
```

In [2]:
```
def f(x):
    return (x + 3) ** 2

def df(x):
    return 2 * (x + 3)
```

In [3]:
```
x = 2
learning_rate = 0.1
precision = 0.000001
max_iterations = 1000
step_size = 1
iteration = 0
x_values = [x]

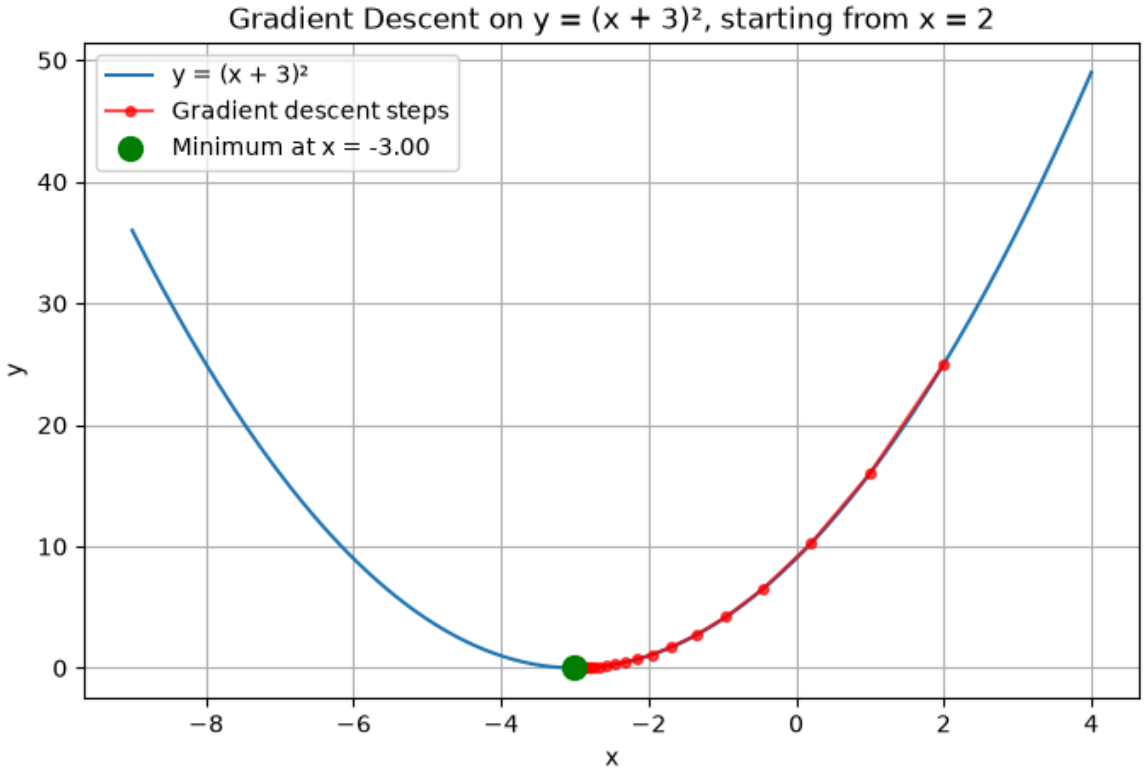
while step_size > precision and iteration < max_iterations:
    previous_x = x
    x = x - learning_rate * df(previous_x)
    step_size = abs(x - previous_x)
    iteration += 1
    x_values.append(x)
    print(f"Iteration {iteration:2d}: x = {x:.6f}, y = {f(x):.8f}")

print()
print("Local minimum occurs at x =", round(x, 4))
print("Minimum value of y =", round(f(x), 6))
print("Number of iterations:", iteration)
```

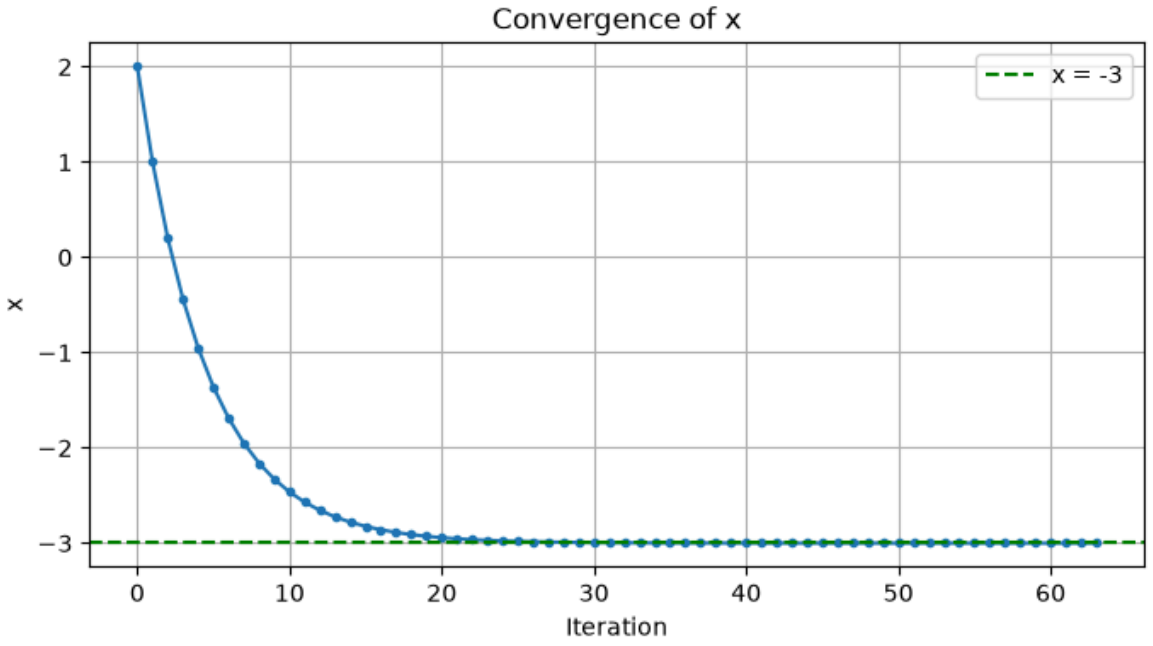
```
Iteration  1: x = 1.000000, y = 16.00000000
Iteration  2: x = 0.200000, y = 10.24000000
Iteration  3: x = -0.440000, y = 6.55360000
Iteration  4: x = -0.952000, y = 4.19430400
Iteration  5: x = -1.361600, y = 2.68435456
Iteration  6: x = -1.689280, y = 1.71798692
Iteration  7: x = -1.951424, y = 1.09951163
Iteration  8: x = -2.161139, y = 0.70368744
Iteration  9: x = -2.328911, y = 0.45035996
Iteration 10: x = -2.463129, y = 0.28823038
Iteration 11: x = -2.570503, y = 0.18446744
Iteration 12: x = -2.656403, y = 0.11805916
Iteration 13: x = -2.725122, y = 0.07555786
Iteration 14: x = -2.780098, y = 0.04835703
Iteration 15: x = -2.824078, y = 0.03094850
Iteration 16: x = -2.859263, y = 0.01980704
Iteration 17: x = -2.887410, y = 0.01267651
Iteration 18: x = -2.909928, y = 0.00811296
Iteration 19: x = -2.927942, y = 0.00519230
Iteration 20: x = -2.942354, y = 0.00332307
Iteration 21: x = -2.953883, y = 0.00212676
Iteration 22: x = -2.963107, y = 0.00136113
Iteration 23: x = -2.970485, y = 0.00087112
Iteration 24: x = -2.976388, y = 0.00055752
Iteration 25: x = -2.981111, y = 0.00035681
Iteration 26: x = -2.984888, y = 0.00022836
Iteration 27: x = -2.987911, y = 0.00014615
Iteration 28: x = -2.990329, y = 0.00009354
Iteration 29: x = -2.992263, y = 0.00005986
Iteration 30: x = -2.993810, y = 0.00003831
Iteration 31: x = -2.995048, y = 0.00002452
Iteration 32: x = -2.996039, y = 0.00001569
Iteration 33: x = -2.996831, y = 0.00001004
Iteration 34: x = -2.997465, y = 0.00000643
Iteration 35: x = -2.997972, y = 0.00000411
Iteration 36: x = -2.998377, y = 0.00000263
Iteration 37: x = -2.998702, y = 0.00000168
Iteration 38: x = -2.998962, y = 0.00000108
Iteration 39: x = -2.999169, y = 0.00000069
Iteration 40: x = -2.999335, y = 0.00000044
Iteration 41: x = -2.999468, y = 0.00000028
Iteration 42: x = -2.999575, y = 0.00000018
Iteration 43: x = -2.999660, y = 0.00000012
Iteration 44: x = -2.999728, y = 0.00000007
Iteration 45: x = -2.999782, y = 0.00000005
Iteration 46: x = -2.999826, y = 0.00000003
Iteration 47: x = -2.999861, y = 0.00000002
Iteration 48: x = -2.999888, y = 0.00000001
Iteration 49: x = -2.999911, y = 0.00000001
Iteration 50: x = -2.999929, y = 0.00000001
Iteration 51: x = -2.999943, y = 0.00000000
Iteration 52: x = -2.999954, y = 0.00000000
Iteration 53: x = -2.999963, y = 0.00000000
Iteration 54: x = -2.999971, y = 0.00000000
Iteration 55: x = -2.999977, y = 0.00000000
Iteration 56: x = -2.999981, y = 0.00000000
Iteration 57: x = -2.999985, y = 0.00000000
Iteration 58: x = -2.999988, y = 0.00000000
Iteration 59: x = -2.999990, y = 0.00000000
Iteration 60: x = -2.999992, y = 0.00000000
Iteration 61: x = -2.999994, y = 0.00000000
Iteration 62: x = -2.999995, y = 0.00000000
Iteration 63: x = -2.999996, y = 0.00000000

Local minimum occurs at x = -3.0
Minimum value of y = 0.0
Number of iterations: 63
```

In [4]:
```
xs = np.linspace(-9, 4, 200)
plt.figure(figsize=(8, 5))
plt.plot(xs, f(xs), label="y = (x + 3)²")
plt.plot(x_values, [f(v) for v in x_values], "o-", color="red", markersize=4, alpha=0.7,
label="Gradient descent steps")
plt.scatter([x], [f(x)], color="green", s=100, zorder=5, label=f"Minimum at x = {x:.2f}")
plt.xlabel("x")
plt.ylabel("y")
plt.title("Gradient Descent on y = (x + 3)², starting from x = 2")
plt.legend()
plt.grid(True)
plt.show()
```



In [5]:
```
plt.figure(figsize=(8, 4))
plt.plot(range(len(x_values)), x_values, marker="o", markersize=3)
plt.axhline(-3, color="green", linestyle="--", label="x = -3")
plt.xlabel("Iteration")
plt.ylabel("x")
plt.title("Convergence of x")
plt.legend()
plt.grid(True)
plt.show()
```



In [6]:
```python
def gradient_descent(start, learning_rate, precision=0.000001, max_iterations=10000):
    x = start
    for i in range(1, max_iterations + 1):
        new_x = x - learning_rate * df(x)
        if abs(new_x - x) < precision:
            return new_x, i
        x = new_x
    return x, max_iterations

for lr in [0.01, 0.05, 0.1, 0.5, 0.9]:
    minimum, steps = gradient_descent(2, lr)
    print(f"Learning rate {lr:<4}: minimum at x = {minimum:.4f} after {steps} iterations")
```

```
Learning rate 0.01: minimum at x = -3.0000 after 571 iterations
Learning rate 0.05: minimum at x = -3.0000 after 126 iterations
Learning rate 0.1 : minimum at x = -3.0000 after 63 iterations
Learning rate 0.5 : minimum at x = -3.0000 after 2 iterations
Learning rate 0.9 : minimum at x = -3.0000 after 73 iterations
```